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2027 CYCLE

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COMEDK UGET

Chemistry

Chemical Bonding & Molecular Structure

Official Syllabus Class 11 · Chapter 4 · COMEDK UGET 2027 Cycle Notes

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Concept Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (COMEDK UGET Chemistry syllabus, Class 11, Chapter 4)	Skills the questions test
Ionic bond	Valence electrons, ionic bond, covalent character of ionic bond.	Electron-transfer logic, lattice-enthalpy ranking, ionic property predictions.
Covalent bond & Lewis structures	Covalent bond, Lewis's structure, valence electrons.	Dot-structure drawing, formal-charge checks, electron-pair counting.
Bond parameters & resonance	Bond parameters, resonance.	Bond length / angle / enthalpy comparisons, bond-order counting, resonance identification.
Polar character of covalent bond	Polar character of covalent bond, covalent character of ionic bond.	Dipole-moment reasoning, Fajan's-rule comparisons.
Geometry & VSEPR	Geometry of covalent molecules, VSEPR theory.	Shape prediction from electron-pair counts, angle compression by lone pairs.
Hybridisation	Concept of hybridization, involving s, p and d orbitals and shapes of some simple molecules.	Hybridisation shortcut calculations, geometry mapping, sigma / pi counting.
Molecular orbital theory	Molecular orbital theory of homonuclear diatomic molecules (qualitative idea only).	Bond order from MO filling, magnetism prediction.
Hydrogen bond	Hydrogen bond.	Types, boiling-point and density consequences.

EXAM PATTERN

COMEDK UGET pattern (2026 information brochure and official syllabus PDFs, the current 2027 baseline): one online paper of 180 questions — Physics 60, Chemistry 60, Mathematics 60 — in 180 minutes, +1 per correct answer and **no negative marking**, so every question should be attempted. The Chemistry section is set at Karnataka PUC / NCERT level, and Chapter 4 supplies the concepts that later chapters (equilibrium, organic chemistry, coordination compounds) build on constantly.

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Why Atoms Bond: the Octet Rule and Its Limits

Look around a room: salt crystals on food, water in a bottle, steel in the fan, plastic in the pen. Every one of these holds together because atoms stick to other atoms in predictable ways — and that stickiness is **chemistry's first question**: why do atoms bond at all, and why in these exact shapes and ratios? Keep one count running in your head through the whole chapter: **where are the electrons, and how many electron pairs sit around the central atom?** Nearly every shape, angle and property answer flows from that single number.

- **Only valence electrons bond:** the outermost-shell electrons take part in bonding; the inner cores stay untouched. Sodium $(2, 8, 1)$ uses just its 1 outer electron; chlorine $(2, 8, 7)$ uses its 7.
- **The octet rule:** atoms bond to reach **eight** electrons in the outer shell — the stable noble-gas configuration ns^2np^6. Neon and argon need no bonds; everything else chases their arrangement.
- **Two routes to the same goal: transfer** electrons (metal + non-metal → **ionic bond**) or **share** electrons (non-metal + non-metal → **covalent bond**). Every structure in this chapter is one of these two moves, or a blend of both.
- **Bonding releases energy:** a bond forms only when the bonded arrangement has **lower** energy than the separated atoms. Energy out is why burning, rusting and setting of cement all involve new bonds.

Drawing Lewis structures: a 6-step routine

Every dot-structure question is the same routine in different clothing. Practise the steps until they run on autopilot:

1. **Count** the total valence electrons — add one per negative charge, remove one per positive charge. CO_2: $4 + 2 \times 6 = 16$ electrons.
2. **Write the skeleton** — the least electronegative atom sits central; hydrogen is always terminal (it can form only one bond).
3. **Join** the atoms with single bonds and complete the octets of the **outer** atoms first.
4. **Park leftovers** on the central atom as lone pairs.
5. **If the central atom is short of an octet**, convert outer lone pairs into double or triple bonds.
6. **Check formal charges** — the best structure carries the smallest formal charges, with any negative charge on the more electronegative atom.

Where the octet rule breaks (and questions begin)

The octet rule is a **guide, not a law** — and its exceptions are exactly where examiners like to plant options. There are three clean families:

- **Incomplete octet:** $BeCl_2$ carries 4 electrons around Be and BF_3 carries 6 around B. Small central atoms accept fewer than 8 and stay stable — these are electron-deficient, which is why BF_3 grabs electron pairs from other molecules aggressively.
- **Expanded octet:** PCl_5 has 10 electrons around P and SF_6 has 12 around S. From the third shell onward, d orbitals open up extra room.

- **Odd-electron molecules:** NO (11 valence electrons) and NO_2 (17) cannot give **every** atom an octet — someone must stay short-handed.

EVERYDAY EXAMPLE — SHARING ONE UMBRELLA

Two friends walking in rain with one umbrella between them **share** it — neither owns it alone, but both stay dry: that is a covalent bond. One friend simply **hands the umbrella over** and both walk dry separately: that is an ionic bond, and the "dryness" each feels is the completed octet. Same umbrella, two arrangements — the chapter is about choosing and predicting the arrangement.

Ionic Bonding: Electrons Transferred

Sodium metal meets chlorine gas and the product is ordinary table salt — a crystalline solid that shatters, melts at $801°C$ and dissolves in water. The whole story is one electron changing owners:

- **Step 1 — donation:** sodium $(2, 8, 1)$ loses its single valence electron and becomes Na^+ with the neon configuration $2, 8$. Losing its one outer electron costs a little energy but the payoff is a full shell.
- **Step 2 — acceptance:** chlorine $(2, 8, 7)$ picks up that electron and becomes Cl^- with the argon configuration $2, 8, 8$.
- **Step 3 — attraction:** the opposite charges pull on each other **from every direction**, so ions stack into a giant three-dimensional crystal — there are no "NaCl molecules", only a $1:1$ **ratio** of ions in the lattice.
- **The force is nondirectional and strong** — which explains almost every ionic property in the table below.

What favours ionic bond formation

Three energy terms decide whether ions form easily, and each is a one-line fact:

- **Low ionization enthalpy** for the metal — the easier it donates, the better. (Caesium and potassium are champion donors.)
- **Strongly negative electron gain enthalpy** for the non-metal — the halogens are eager acceptors.
- **High lattice enthalpy** — the crystal glue that repays the energy spent on ion formation. Bigger charges and smaller ions mean a more negative (stronger) lattice enthalpy.

LATTICE ENTHALPY

Lattice enthalpy = energy released when 1 mol of an ionic crystal forms from gaseous ions

Trends: charge ↑↑ $\Longrightarrow$ lattice enthalpy ↑↑ ion radius ↑ $\Longrightarrow$ lattice enthalpy ↓

Read the trend as physics: double the charges and the attraction roughly **quadruples** (Coulomb's law), which is why $2+/2-$ salts beat $1+/1-$ salts by a wide margin. Shrink the ions and the charges sit closer together, again strengthening the pull. This single idea answers every "arrange by melting point / lattice enthalpy / hardness" question on ionic solids.

EVERYDAY EXAMPLE — MAGNETS ON A FRIDGE DOOR

Tiny strong magnets packed close together clamp far harder than weak magnets held far apart — small ionic radii and high charges do exactly this inside a crystal. Pull one magnet off the sheet and its neighbours barely care: that brittleness is why a salt crystal **cleaves and shatters** instead of bending, because shifting one layer lines like charges up against like charges.

Properties of ionic compounds — cause and effect

Property	Why (the cause behind it)
High melting and boiling points	Electrostatic attraction acts in all directions throughout the crystal; large energy is needed to pull the layers apart
Brittle	Shifting a layer brings like charges face to face — repulsion cracks the crystal along the plane
Conduct in molten / aqueous state only	Ions must be free to move; in the solid they are locked in place (in water or melt, the lattice opens and ions travel)
Soluble in polar solvents (water)	Water molecules surround and pull off ions, paying back some lattice enthalpy with hydration energy
Do not conduct as solids	Electrons are localized; charge carriers exist but cannot migrate

WORKED EXAMPLE — RANKING LATTICE ENTHALPIES

Rank NaCl, MgO and CsCl by lattice enthalpy and predict which has the highest melting point.

Solution. Compare charges first, radii second. MgO has $2\frac{+}{2}-$ ions; NaCl and CsCl have $1\frac{+}{1}-$. Doubling both charges makes the attraction about four times stronger, so MgO dominates. Between the two $1\frac{+}{1}-$ salts, Na^+ is smaller than Cs^+, so NaCl packs the charges closer:

$$\text{MgO} \gg \text{NaCl} > \text{CsCl} \quad \text{in lattice enthalpy, and in the same order for melting point}$$

Measured melting points confirm the ranking: MgO $2852°C$, NaCl $801°C$, CsCl $645°C$ — the trend rule did all the work before any data appeared.

Covalent Bonding: Electrons Shared

Two hydrogen atoms each bring one electron; by **sharing the pair**, each experiences a full $1s^2$ shell part of the time — and the shared pair spends its time **between** the two nuclei, gluing them together. That is the whole covalent idea, scaled up to millions of molecules: electrons **shared between** atoms instead of **transferred between** them.

Formal charge: which Lewis structure wins

When several dot structures are possible, formal charge picks the best one. It is bookkeeping, not real charge — but the bookkeeping has rules:

FORMAL CHARGE

$$\text{FC} = \text{valence e}^- - \text{non-bonding e}^- - \frac{1}{2}(\text{bonding e}^-) \quad = \quad V - L - \frac{B}{2}$$

- **Smallest formal charges wins:** the structure with the fewest and smallest charges is the major contributor.
- **Negatives go on electronegative atoms:** if a negative formal charge must exist, it belongs on the atom that pulls electrons hardest.
- **Never skip the check:** structures that "look fine" can carry absurd charges like +2 on oxygen — the formal-charge screen catches them instantly.

WORKED EXAMPLE — OZONE'S TWO FACES

Ozone, O_3, has 18 valence electrons and two possible skeletons with one double bond and one single bond. Compute the formal charges of the central oxygen and the two end oxygens.

Solution. Central O: 2 bonds (4 bonding electrons) + 1 lone pair, so $\text{FC} = 6 - 2 - \frac{4}{2} = 0$. The double-bonded end O: $\text{FC} = 6 - 4 - \frac{4}{2} = 0$. The single-bonded end O: $\text{FC} = 6 - 6 - \frac{2}{2} = -1$. So the molecule carries a -1 on one end and $+1$ on the central atom in **either** single structure — both structures are equal partners, which is exactly why real ozone is the **resonance hybrid** below: both bonds identical, each roughly 1.5 bonds.

Bond parameters: measuring a bond

Four numbers describe any bond, and each one answers a different question:

- **Bond length:** distance between nuclei of the bonded atoms, measured in picometres ($1\ \text{pm} = 10^{-12}\ \text{m}$). Shorter → stronger, usually.
- **Bond angle:** the angle between two bonds at the same atom — the shape fingerprint (H_2O 104.5°, CH_4 109.5°).
- **Bond enthalpy:** energy to break one mole of that bond in the gas phase — the strength meter.
- **Bond order:** the number of shared pairs between two atoms — 1 for single, 2 for double, 3 for triple.

Bond	Length (pm)	Enthalpy (kJ/mol)	Order
H – H	74	436	1
Cl – Cl	199	242	1
O – H	96	463	1

Bond	Length (pm)	Enthalpy (kJ/mol)	Order
C – C	154	348	1
C = C	134	614	2
C ≡ C	120	839	3

Read the carbon rows as the master trend: as bond order climbs, length shrinks and enthalpy climbs — more shared pairs pull the nuclei closer and cost more to break. Any "arrange by bond length / strength" question starts by counting bond orders, not by memorising numbers.

Resonance: one real structure between the drawings

When a single Lewis structure cannot represent a molecule, the real structure is a **hybrid** of several valid drawings — the molecule **is** the average, it does not "flip" between them.

- **Ozone:** both bonds are equal at 128 pm — longer than a double bond (110 pm), shorter than a single bond (148 pm). Two drawn structures, one real molecule.
- **Carbonate, CO_3^{2-}:** all three C–O bonds are identical at about 129 pm; the −2 charge spreads over all three oxygens equally.
- **Delocalisation stabilises:** the hybrid is always more stable than any single drawn structure — the extra stability is called **resonance stabilisation**.
- **Not switching:** the double bond does not "jump around"; the electrons are spread over the whole framework all the time.

EVERYDAY EXAMPLE — A CEILING FAN PHOTO

Photograph a spinning fan and the blade appears everywhere along its arc at once — the photo is not wrong, it is an **average of positions**. A resonance hybrid is that photo: the electrons genuinely sit spread over several positions simultaneously, and each Lewis structure is just one frozen frame.

Polarity, dipole moment and Fajan's rules

When two **different** atoms share electrons, the sharing is unequal: the more electronegative atom pulls the pair closer and turns slightly negative (δ^-), leaving the other slightly positive (δ^+). The bond becomes a tiny magnet — a **dipole**.

DIPOLE MOMENT

$$\mu = q \times d \quad \text{measured in Debye (D)} \qquad \mu_{\text{net}} = \text{vector sum of all bond dipoles}$$

- **Vector, not arithmetic:** bond dipoles **add as arrows**. Symmetric molecules cancel their own arrows and end at zero.
- **The classics:** CO_2 (linear) has two equal, opposite dipoles → net 0. H_2O (bent) has two dipoles that do not cancel → net 1.85 D. BF_3 (trigonal planar) and CH_4 (tetrahedral) cancel perfectly → net 0.

- **NH_3 beats NF_3:** both are pyramidal, yet $\mu(NH_3) = 1.47$ D $> \mu(NF_3) = 0.23$ D. In ammonia the N–H dipoles and the lone pair point the **same** way; in nitrogen trifluoride the strong N–F dipoles point **against** the lone pair and nearly cancel it.
- **Covalent character in ionic bonds (Fajan's rules):** a **small, highly charged** cation distorts (polarises) a **large** anion's electron cloud, making the "ionic" bond partly covalent. Small cation + big anion + high charges → more covalent character.

WORKED EXAMPLE — WHY ALCL3 BEHAVES COVALENT

Explain why $AlCl_3$ is far more covalent in character than NaCl, though both are metal chlorides.

Solution. Apply Fajan's rules to the cation: $|Al^{3+}|$ is smaller **and** carries triple the charge of Na^+. Its intense positive core drags hard on the chloride cloud, deforming it deeply toward the aluminium — shared-electron behaviour inside a formally ionic formula. Na^+ is large and singly charged, so Cl^- keeps its round shape: essentially ionic. The ranking $|NaCl < MgCl_2 < AlCl_3|$ in covalent character follows from charge and size alone.

VSEPR: Predicting Molecular Shapes

Electron pairs around a central atom carry negative charges — they **repel** — so they arrange themselves as far apart as the geometry allows. That single idea is **VSEPR** (valence shell electron pair repulsion), and it predicts shapes without any calculation: count the electron domains, spread them out, then **subtract** the visual effect of lone pairs.

- **The repulsion ranking:** lone pair–lone pair > lone pair–bond pair > bond pair–bond pair. A lone pair is held by one nucleus only, so it spreads wider and squeezes the bonds harder.
- **Lone pairs compress angles:** each lone pair on a tetrahedral set-up steals roughly 2° – 3°: CH_4 109.5° → NH_3 107° → H_2O 104.5°.
- **Count domains, not just atoms:** a double bond counts as **one** domain. CO_2 has two domains → linear, even though it has two double bonds.

The 3-step prediction routine

1. **Draw** the Lewis structure; count **all** electron domains around the central atom (bonds — single, double, triple alike — plus lone pairs).
2. **Assign** the parent geometry: 2 domains linear, 3 trigonal planar, 4 tetrahedral, 5 trigonal bipyramidal, 6 octahedral.
3. **Place** lone pairs and name the **molecular** shape they leave behind — lone pairs in trigonal bipyramids always take equatorial spots first, where they feel less crowding.

Type	Electron geometry	Bond angle(s)	Molecular shape	Example
AB_2	Linear	180°	Linear	$BeCl_2$, CO_2
AB_3	Trigonal planar	120°	Trigonal planar	BF_3

Type	Electron geometry	Bond angle(s)	Molecular shape	Example
AB_2E	Trigonal planar	$< 120°$	Bent	SO_2, $SnCl_2$
AB_4	Tetrahedral	$109.5°$	Tetrahedral	CH_4
AB_3E	Tetrahedral	$107°$	Trigonal pyramidal	NH_3
AB_2E_2	Tetrahedral	$104.5°$	Bent	H_2O
AB_5	Trigonal bipyramidal	$120°$, $90°$	Trigonal bipyramidal	PCl_5
AB_4E	Trigonal bipyramidal	$< 120°$, $< 90°$	See-saw	SF_4
AB_3E_2	Trigonal bipyramidal	$90°$	T-shaped	ClF_3
AB_2E_3	Trigonal bipyramidal	$180°$	Linear	XeF_2, I_3^-
AB_6	Octahedral	$90°$	Octahedral	SF_6

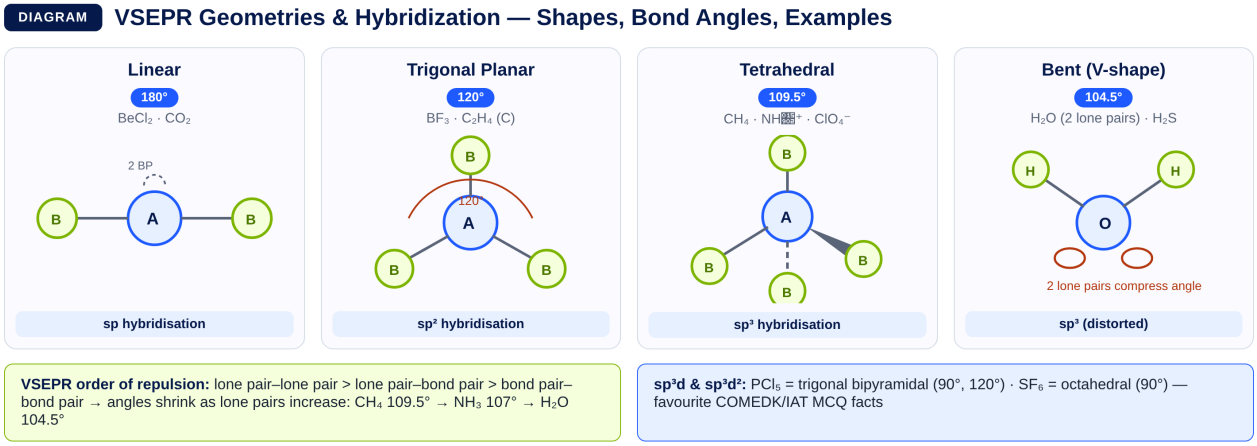


Figure 1 — The four most-tested VSEPR shapes with their bond angles, hybridisation and examples.

EVERYDAY EXAMPLE — BALLOONS TIED TOGETHER

Tie two balloons and they stand opposite each other; three spread into a triangle; four push into a tetrahedron; five and six build the bipyramid and octahedron. Balloons repel like electron pairs do — same number of balloons, same shape. Now squeeze one balloon harder (a fatter lone pair) and the others get pushed closer: that is angle compression in NH_3 and H_2O.

WORKED EXAMPLE — CHLORINE TRIFLUORIDE

Predict the shape of ClF_3: chlorine brings 7 valence electrons, bonds to three fluorines.

Solution. Count domains: 3 bond pairs + 2 lone pairs = 5 domains → trigonal-bipyramidal parent. The 2 lone pairs take **equatorial** positions, leaving the two axial and one equatorial fluorine: a **T-shaped** molecule with bonds near 90°. The same routine gives see-saw for SF_4 (4 bonds + 1 lone pair) and linear XeF_2 (2 bonds + 3 lone pairs) — one routine, every sp^3d family answer.

Valence Bond Theory and Hybridisation

Valence bond theory (VBT) explains the covalent bond as **orbital overlap**: two half-filled orbitals merge, and the shared electron density between the nuclei cements them. More overlap → stronger bond — which is why head-on overlap beats sideways overlap, and why bonds have the two distinct personalities below:

Feature	Sigma (σ) bond	Pi (π) bond
Overlap style	Head-on, along the internuclear axis	Sideways, above and below the axis
Strength	Stronger (maximum overlap)	Weaker (partial overlap)
Free rotation	Allowed — atoms can spin about the bond	Blocked — rotation breaks the overlap (source of cis–trans isomers)
Occurrence	Every single, double and triple bond has exactly one	Only in double and triple bonds, alongside the sigma

- **Counting rule:** a molecule with n multiple bonds has one sigma per bond pair and one extra pi per double bond (two per triple). Ethene C_2H_4: 5 sigma + 1 pi. Ethyne C_2H_2: 3 sigma + 2 pi.

Hybridisation: why methane is tetrahedral, not square

Free carbon has **two** unpaired p electrons, which would predict CH_2 with a 90° bond — but methane has four identical bonds at 109.5°. The fix: just before bonding, one $2s$ electron promotes to the empty $2p$, and the one s + three p orbitals **blend into four equivalent sp^3 hybrids**, directed at the corners of a tetrahedron. Hybridisation is **bookkeeping before bonding** — real orbitals of slightly different energy are replaced by identical hybrids pointing where the geometry needs them.

- **Number is conserved:** mixing n atomic orbitals always produces n hybrids — one s + three p gives four sp^3, never more.
- **Hybrids point where pairs sit:** lone pairs occupy hybrids too — they simply are not bonds, which is why NH_3 is still sp^3 but pyramidal.
- **Unhybridised p orbitals build pi bonds:** after sp^2 mixing, the leftover p orbital sticks out sideways and forms the pi bond of ethene.

THE HYBRIDISATION SHORTCUT

$$H = \frac{1}{2}(V + M - C + A)$$

V = valence electrons of the central atom M = monovalent atoms bonded

C = cation charge on the species A = anion charge on the species

$H = 2 \to \text{sp}$ $H = 3 \to \text{sp}^2$ $H = 4 \to \text{sp}^3$

$H = 5 \to \text{sp}^3\text{d}$ $H = 6 \to \text{sp}^3\text{d}^2$

Count lone pairs too — they live in hybrids, they just form no bonds.

Hybrid	Geometry	Angle	Examples
sp	Linear	180°	$BeCl_2$, CO_2 (carbon), C_2H_2
sp^2	Trigonal planar	120°	BF_3, SO_2 (shape bent), C_2H_4
sp^3	Tetrahedral	109.5°	CH_4, NH_3 (pyramidal), H_2O (bent), NH_4^+
sp^3d	Trigonal bipyramidal	120°, 90°	PCl_5, SF_4 (see-saw), ClF_3 (T-shape)
sp^3d^2	Octahedral	90°	SF_6, XeF_4 (square planar)

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WORKED EXAMPLE — HYBRIDISATION ON AUTOPILOT

Find the hybridisation and shape of (a) SF_4, (b) NH_4^+, (c) the carbons in C_2H_4.

Solution. (a) $H = \frac{1}{2}(6 + 4) = 5 \to \text{sp}^3\text{d}$; one of the five pairs is a lone pair, so the shape left is **see-saw**. (b) $H = \frac{1}{2}(5 + 4 - 1) = 4 \to \text{sp}^3$ with **no** lone pair: perfect **tetrahedral**, exactly like methane. (c) Each carbon has 3 sigma bonds (two H + one C) and **no** lone pair $\to \text{sp}^2$, planar, with the leftover p orbitals making the pi bond. Every hybridisation question is this two-line count.

Molecular Orbital Theory: Bonding across the Whole Molecule

VBT localises electrons between two atoms; **molecular orbital theory (MOT)** goes further — atomic orbitals of **all** atoms combine into molecular orbitals that belong to the whole molecule. For two identical atoms (homonuclear diatomics), each pair of atomic orbitals makes two MOs, and the syllabus needs only the qualitative picture:

- **In-phase combination → bonding MO:** electron density piles **between** the nuclei; energy drops; the bond holds.
- **Out-of-phase combination → antibonding MO ($\star$):** a node between the nuclei; energy rises; electrons here **weaken** the bond.

- **Two atomic orbitals → two MOs:** nothing is created or destroyed, only reorganised.
- **Fill from the bottom up**, exactly like atomic orbitals — Aufbau, then Hund.

Filling order and bond order

For the second-row diatomics up to nitrogen, s–p mixing pushes $\pi 2p$ **below** $\sigma 2p$; from oxygen onward the textbook order applies. The count that matters is:

BOND ORDER (MOT)

$$\text{Bond order} = \frac{N_b - N_a}{2} \qquad N_b = \text{electrons in bonding MOs}, \quad N_a = \text{electrons in antibonding MOs}$$

$$\text{Bond order} > 0 \implies \text{the molecule exists; higher bond order} \rightarrow \text{shorter, stronger bond}$$

Species	Key filling	Bond order	Magnetism	Takeaway
H_2	2 e^- in $\sigma 1s$	1	Diamagnetic	Simplest stable bond
He_2	$\sigma 1s$ full and $\sigma^* 1s$ full	0	—	Does not exist — MOs cancel the bond
N_2	$\pi 2p$ level fully filled	3	Diamagnetic	Strongest common diatomic bond
O_2	2 unpaired e^- in $\pi^* 2p$	2	**Paramagnetic**	MOT's headline success
F_2	π^* filled	1	Diamagnetic	Weakest of the trio

THE O2 MOMENT — WHERE LEWIS FAILS

The Lewis structure of O_2 shows a double bond with **all electrons paired**, predicting diamagnetism. Liquid oxygen in a magnet field **sticks to the poles** — it is paramagnetic, and only MOT explains it: the two highest electrons occupy the degenerate $\pi^* 2p$ orbitals singly (Hund's rule), giving 2 unpaired spins. Whenever a question pairs "double bond" with "paramagnetic", the answer is the antibonding pair — no other model in Class 11 produces it.

WORKED EXAMPLE — N2 VS N2+

N_2 has 14 electrons; remove one to form N_2^+. Compare their bond orders and predict which is more stable.

Solution. In N_2: $N_b = 10$, $N_a = 4$, so bond order $= \frac{10-4}{2} = 3$. The removed electron comes from a **bonding** $\pi 2p$ orbital (the highest filled level for nitrogen), so N_2^+ has $N_b = 9$, $N_a =$

4: bond order $= \frac{5}{2} = 2.5$. Lower bond order → longer, weaker bond: N_2 is more stable. The same bookkeeping for O_2^+ removes an **antibonding** electron and **raises** bond order to 2.5 — always check **which** MO loses the electron.

Hydrogen Bonding: The Bond That Isn't Quite a Bond

When hydrogen is bonded to a **small, highly electronegative** atom — fluorine, oxygen or nitrogen — the shared electrons drift so far toward the partner that H becomes almost a bare proton. That δ^+ hydrogen then attracts the lone pair of a neighbouring F, O or N atom: an electrostatic handshake called the **hydrogen bond** — weaker than a covalent bond (about 10–40 kJ/mol versus 400 kJ/mol) but far stronger than ordinary attractions between molecules.

- **Two requirements, both necessary:** H bonded directly to F, O or N, and a lone pair on the neighbouring molecule to receive it.
- **Intermolecular:** between separate molecules — water chains, HF zig-zags, ammonia clusters.
- **Intramolecular:** within one molecule — ortho-nitrophenol's OH hydrogen reaches the neighbouring $-NO_2$ oxygen on the same ring.
- **Direction matters:** intramolecular H-bonding **shields** the molecule from clustering with neighbours — ortho-nitrophenol is steam-volatile while the para isomer (no intramolecular option) is not.

What hydrogen bonding explains (classic answers)

- **Water boils at 100 °C** while H_2S boils at $-60°C$ — oxygen's stronger pull builds stronger H-bonds, and extra energy is needed to free the molecules.
- **Ice floats:** H-bonds lock water into an open hexagonal cage with empty space; when ice melts, the cage partly collapses and molecules pack closer — liquid water is **denser** than ice, a rarity among substances.
- **HF is a weak acid:** the H-bond network holds the released H^+ back, damping ionisation.
- **Ammonia's high solubility and high boiling point** for its size — same cause.

EVERYDAY EXAMPLE — VELCRO STRAPS

Velcro grips firmly, yet opens and re-closes thousands of times: strong enough to hold shape, weak enough to rearrange. Hydrogen bonds behave identically in water — constantly breaking and reforming as molecules slide past each other (liquid flows), but grabbing hard when pulled apart (high boiling point, high surface tension). Water's personality **is** hydrogen bonding's personality.

WORKED EXAMPLE — BOILING POINT RANKING

Rank H_2O, HF and NH_3 by boiling point using hydrogen bonding.

Solution. Count the H-bond **handshakes per molecule**. Water donates two H atoms and accepts two more through two lone pairs — up to four links, a dense 3-D network. HF has

a stronger single bond but only one donor and three lone pairs; ammonia donates three but accepts only one. Net result:

$$H_2O\ (100°C) > HF\ (19.5°C) > NH_3(-33°C)$$

Counting links beats judging single-bond strength — the network, not one bond, sets the boiling point.

Formula Cheatsheet — Revise the Chapter in One Glance

FORMULA CHEATSHEET

Item	Relation	Use it for
Formal charge	$FC = V - L - \frac{B}{2}$	Choosing the best Lewis structure
Bond order (Lewis)	shared pairs between the two atoms	Length / enthalpy trend questions
Dipole moment	$\mu = q \times d$, vector sum over all bonds	Polarity of bent vs linear molecules
Fajan's rules	small cation + big anion + high charges $\rightarrow$ covalent character	"Which ionic bond is more covalent"
Hybridisation shortcut	$H = \frac{1}{2}(V + M - C + A)$	Instant sp / sp^2 / sp^3 / sp^3d / sp^3d^2
MO bond order	$\frac{N_b - N_a}{2}$	Stability, magnetism, ion comparison
Repulsion order	lp–lp > lp–bp > bp–bp	Angle compression in NH_3, H_2O
Bond-order trend	order $\uparrow \rightarrow$ length $\downarrow$, enthalpy $\uparrow$	C–C vs C=C vs C≡C rankings
Lattice enthalpy	charge $\uparrow\uparrow$ and radius $\downarrow \rightarrow$ lattice enthalpy $\uparrow$	Melting-point rankings of salts
H-bond strength ladder	covalent (approx. 400 kJ/mol) >> H-bond (10–40) > van der Waals	Explaining bp / density anomalies

The Whole Chapter on One Page

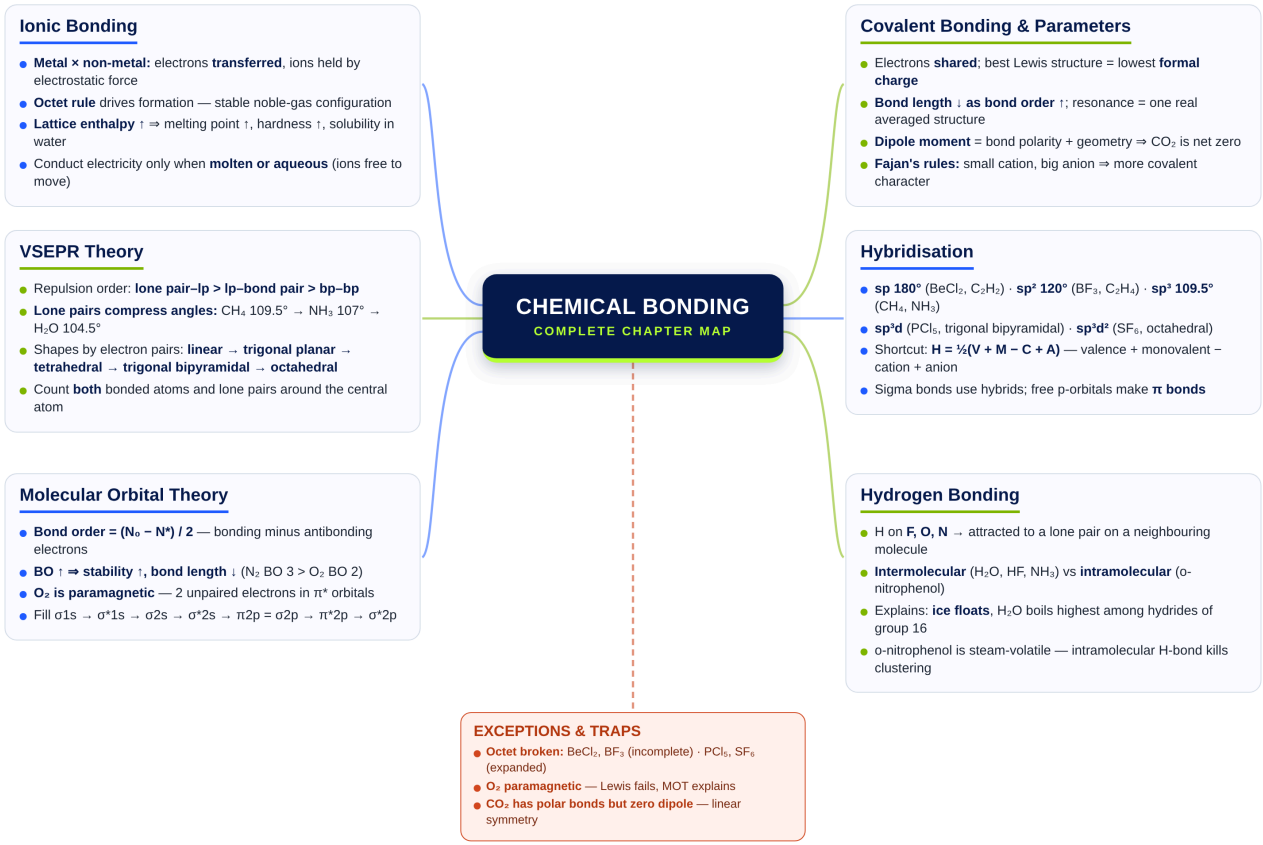


Figure 2 — Chemical Bonding mindmap: every branch of the chapter with its headline facts, plus the exception corner.

Use this map as the revision ritual: cover each branch, recite its four leaves, and mark the ones that stumble. The exceptions card at the bottom is the highest-value minute of revision the night before the exam — those six lines are the difference between two very similar-looking options.

Exceptions & Traps — Where Questions Hide

EXCEPTIONS & TRAPS

Rule you trust	The exception	The trap to dodge
Octet rule: 8 electrons always	$BeCl_2$ (4), BF_3 (6) incomplete; PCl_5 (10), SF_6 (12) expanded; NO, NO_2 odd	Forcing a double bond onto Be or B to "complete" the octet
All double bonds mean diamagnetic	O_2 is paramagnetic (2 unpaired π^* electrons)	Answering from the Lewis structure — MOT decides magnetism

Rule you trust	The exception	The trap to dodge
Polar bonds → polar molecule	CO_2, BF_3, CH_4 have zero net dipole	Adding dipoles arithmetically instead of as vectors
NF_3 should beat NH_3 in dipole (F pulls hard)	NH_3 1.47 D > NF_3 0.23 D	Forgetting the lone pair's arrow opposes N–F dipoles
Removing an electron always weakens a bond	O_2^+ has **higher** bond order than O_2	Not checking whether the lost electron was antibonding
Solids sink in their own liquid	Ice floats — H-bond cage keeps water 9% less dense	Forgetting the open cage structure of ice
Isomers have identical volatility	ortho-nitrophenol is steam-volatile, para is not	Ignoring intramolecular vs intermolecular H-bonding

Practice Questions — COMEDK Pattern (10 MCQs)

Attempt these under exam timing: about 60–90 seconds per question. Fully worked solutions follow, so commit to an answer before reading on. Every question maps to one of the sections above.

Q1. Which of the following solids has the highest lattice enthalpy?

(A) NaCl **(B)** KCl
(C) MgO **(D)** CsCl

Q2. The number of sigma and pi bonds in ethene (C_2H_4) is:

(A) 4 σ and 2 π **(B)** 5 σ and 1 π
(C) 6 σ and 0 π **(D)** 3 σ and 3 π

Q3. The shape of the SF_4 molecule is:

(A) Tetrahedral **(B)** See-saw
(C) Square planar **(D)** Linear

Q4. Which molecule is paramagnetic?

(A) N_2 **(B)** O_2
(C) F_2 **(D)** CO_2

Q5. The bond order of N_2^+ is:

(A) 3 **(B)** 2.5
(C) 2 **(D)** 1.5

Q6. Which of these molecules has zero dipole moment?

(A) H_2O **(B)** NH_3
(C) BF_3 **(D)** HCl

Q7. The hybridisation of phosphorus in PCl_5 is:

(A) sp^2 **(B)** sp^3
(C) sp^3d **(D)** sp^3d^2

Q8. Which species has the shortest bond length?

(A) O_2^+ **(B)** O_2
(C) O_2^- **(D)** O_2^{2-}

Q9. ortho-Nitrophenol is steam-volatile while para-nitrophenol is not, because ortho-nitrophenol has:

(A) Stronger covalent bonds **(B)** Intramolecular hydrogen bonding
(C) Intermolecular hydrogen bonding **(D)** A lower molecular mass

Q10. By Fajan's rules, the most covalent in character among these is:

(A) NaCl **(B)** $MgCl_2$
(C) $AlCl_3$ **(D)** $SiCl_4$ (treated as ionic comparison of cations)

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Detailed Solutions

Sol. Q1. Lattice enthalpy grows with ionic charge and shrinks with ionic radius. MgO pairs $2+$ with $2-$ charges — roughly four times the attraction of the $1\tfrac{+}{1}-$ salts. Ranking: MgO $\gg$ NaCl $>$ KCl $>$ CsCl (radius grows down the group).

Answer: (C)

Sol. Q2. Ethene's skeleton: 4 C–H bonds + 1 C–C sigma bond = 5 sigma; the second pair of the C=C is the single pi bond (leftover p orbitals overlap sideways). Formula check: one sigma per bond pair, one extra pi per double bond.

Answer: (B)

Sol. Q3. Sulfur brings 6 valence electrons: 4 bond with F, leaving 1 lone pair. Domains $= 4 + 1 = 5 \rightarrow sp^3d$ parent (trigonal bipyramid). The lone pair takes an equatorial slot, leaving a see-saw shape.

Answer: (B)

Sol. Q4. MOT fills O2′s two degenerate π^*2p orbitals with one electron each (Hund's rule) — 2 unpaired spins → paramagnetic. N_2 and F_2 have all electrons paired; CO_2 is a closed-shell, linear, diamagnetic molecule.

Answer: (B)

Sol. Q5. For N_2: $N_b = 10$, $N_a = 4$, bond order 3. Ionisation removes one electron from the **bonding** $\pi 2p$ level: bond order $= \frac{9-4}{2} = 2.5$.

Answer: (B)

Sol. Q6. BF_3 is trigonal planar and symmetric — the three B–F dipoles cancel exactly (net 0). H_2O (bent), NH_3 (pyramidal) and HCl (two atoms only) all carry net dipoles.

Answer: (C)

Sol. Q7. Shortcut: $H = \frac{1}{2}(V + M - C + A) = \frac{1}{2}(5 + 5) = 5 \rightarrow$ sp^3d, trigonal-bipyramidal parent — the classic 10-electron expanded octet.

Answer: (C)

Sol. Q8. Bond order decides length. O_2 has order 2; O_2^+ loses an **antibonding** electron → order 2.5; O_2^- and O_2^{2-} **add** antibonding electrons → orders 1.5 and 1. Highest order wins: O_2^+ is shortest.

Answer: (A)

Sol. Q9. In the ortho isomer, the OH hydrogen reaches the $-NO_2$ oxygen on the **same** ring — intramolecular H-bonding that satisfies the molecule internally, so it clusters weakly with neighbours and vaporises easily with steam. The para isomer can only bond **between** molecules, so it holds together as a solid.

Answer: (B)

Sol. Q10. Covalent character rises as the cation gets smaller and more highly charged (Fajan's rules): Na^+ (large, $1+$) → Mg^{2+} (smaller, $2+$) → Al^{3+} (smallest, $3+$). Aluminium chloride polarises the chloride cloud most strongly.

Answer: (C)

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